

XIV. *Poisson's and Green's Theorems in Riemann's n -manifold both when it is space-like and when time-space-like.* By Prof. ALEX. MCAULAY, M.A., *University of Tasmania* *.

THE notations of this paper had been used in two other papers sent to the Phil. Mag. a little earlier than the present paper and were not again explained. As, however, these two papers have never appeared, it may now be said (May 1923) that the notations will be found in "Multenions and Differential Invariants," I., II., III. (Proc. Roy. Soc., A. vol. xcix. p. 292, vol. cii. p. 210, and some vol. still unknown).

§ 1. *Scope of the paper and enunciation of the two principal theorems.*—In this § 1 Poisson's theorem is enunciated for each of the two species of Riemann's n -manifold. In § 2 a minute explanation of the meaning of the enunciations is given. The opportunity is used to present several aspects of matters affecting the theorems, which are at the same time of general import. The proof follows in § 3, and the paper concludes with a treatment of Green's theorem for the case of the n -manifold.

In a space-like manifold—that is, one all of whose n -dimensions are real—Poisson's theorem becomes

$$\left. \begin{aligned} V_0 \nabla (\bar{\eta} \nabla \int^n h \bar{m} db) &= \pi_n \bar{m}, \\ \text{where } h &= (\text{distance})^{2-n}/(n-2), \end{aligned} \right\} \dots \dots (1)$$

except when $n=2$, in which case $h = \log (\text{distance})$.

In a time-space-like manifold—that is, a manifold of which one dimension is time-like or imaginary and the remaining $n-1$ are space-like or real—the theorem becomes

$$\left. \begin{aligned} V_0 \nabla (\bar{\eta} \nabla \int^{n-1} h \bar{m} V_0 \iota d\alpha) &= \pi_{n-1} \bar{m}, \\ \text{where } h &= (2z + V_0^2 \iota \nabla z)^{\frac{3-n}{2}} / (n-3), \quad z = \frac{1}{2} (\text{distance})^2, \end{aligned} \right\} \dots (2)$$

except when $n=3$, in which case

$$(n=3), \quad h = \frac{1}{2} \log (2z + V_0^2 \iota \nabla z). \dots \dots (3)$$

It may be stated also that, putting

$$v = N \nabla z = V_1 \iota V_2 \eta \iota \nabla z, \dots \dots (4)$$

h may alternatively be defined by

$$h = (-V_0 v \eta^{-1} v)^{\frac{3-n}{2}} / (n-3), \dots \dots (5)$$

except when $n=3$ as before.

* Communicated by the Author.

It must be explained that

$$\pi_2 = 2\pi, \quad \pi_3 = 4\pi, \quad \pi_4 = 2\pi^2,$$

and generally π_n is the solid angle subtended at a point by the whole of an n -manifold when the manifold is space-like. [The similar solid angle in a time-space-like manifold, but subtended by the whole of the manifold within either light semi-cone, or subtended by the whole without the cone, is in both cases infinite.] If k is any positive integer, the following are the values of π_n [except for π_2 , though the formula is easily modified to cover the case by writing $2^{k-1}(k-1)!$ for the denominator on the right of the first of the following equations]:

$$\left. \begin{aligned} \pi_{2k} &= (2\pi)^k / [2 \cdot 4 \dots (2k-2)], \\ \pi_{2k+1} &= 2(2\pi)^k [1 \cdot 3 \dots (2k-1)]. \end{aligned} \right\} \dots \dots (6)$$

Here is a table for π_n , calculated by 5-figure logarithms, so that last-figure accuracy is not guaranteed:—

n .	π_n .	n .	π_n .	n .	π_n .	n .	π_n .
2	6·2832	8	32·470	14	8·3898	20	·51614
3	12·566	9	29·687	15	5·7217	21	·29294
4	19·739	10	25·502	16	3·7653	22	·16215
5	26·319	11	20·725	17	2·3967	23	·08765
6	31·006	12	16·023	18	1·4786	24	·04631
7	33·074	13	11·838	19	·88582	25	·02394

π_n rapidly diminishes beyond the limits of the table; $\pi_{100} = 10^{-35} \cdot 2 \cdot 3683$. Note that a maximum of about 33 is reached at $n=7$.

π_n is defined as the area of a small sphere, radius r , in an n -dimensional space, divided by r^{n-1} . Thus the area and volume of the sphere are $\pi_n r^{n-1}$ and $\pi_n r^n/n$ respectively, and the circumference and area of a circle of the same radius in the same space are $\pi_{n-1} r^{n-2}$ and $\pi_{n-1} r^{n-1}/(n-1)$, respectively. The volume of the sphere is generated by the motion of a circle of varying radius, which moves parallel to itself and whose radius is $r \cos \theta$ when it is at a distance $r \sin \theta$ from the centre of the sphere. Hence we have

$$\pi_n/n = (\pi_{n-1}/(n-1)) \cdot 2 \int_0^{\frac{1}{2}\pi} (\cos \theta)^n d\theta.$$

From this we further deduce that

$$\pi_{n+2}/\pi_n = 2\pi/n,$$

and (6) then readily follows.

§ 2. *The meaning of the theorems explained.*—Take ρ for
L 2

position vector, and η for the fundamental covariant vector linity, so that the scalar form is $V_0 d\rho \eta d\rho$ (though in a space-like manifold, except when consistency forbids, it is better to reverse the sign of η). Whenever a bar, as with $\bar{\eta}$ and $\bar{m}$ in (1) and (2) of § 1, is placed over a symbol it is understood that the symbol is an invariant, covariant, or contravariant quantity multiplied by k , where $k = |\eta|^{\frac{1}{2}}$ and $|\eta|$ stands for the determinant of η . Such a barred symbol is called a density. [Will somebody invent a better word?—this clashes confusingly with density as applied to matter.] Thus $\bar{\eta}$ is defined to mean the frequently recurring linity $k\eta^{-1}$. $\bar{m}$ is k multiplied by a scalar invariant function of position; it appears as the analogue below of material density when Poisson's theorem is taken in its strictly original sense.

(1), so obviously like the original Poisson theorem, serves to introduce (2). Let, in the case of (1), $r (=T\rho)$ be the distance of the point ρ from the origin O, whether of three-dimensional Euclidean space or of n -dimensional Riemannian. In the three-dimensional case, if we put $h = r^{-1}$ [see (1)] and form the volume integral $\iiint h m db$ for any given volume, including the origin, Poisson's equation asserts that

$$4\pi\bar{m} = \nabla^2 \iiint h \bar{m} db = S \nabla (\nabla \iiint h \bar{m} db),$$

the $\bar{m}$ on the left meaning the value at the origin. In the general Riemannian case h becomes $r^{2-n}/(n-2)$, where, when we are using equiradial coordinates, r may still be defined as $T\rho$, but in general must be defined as distance (along a geodesic) between the element db and the origin. The operator $\nabla^2 = S \nabla \nabla$ becomes the invariant operator

$$V_0 \nabla (\bar{\eta} \nabla [\])$$

(not that the result is an invariant, but that it is an invariant density), which for the future we shall write as $V^2 \nabla \bar{\eta} \nabla$. The 4π becomes π_n , and the equation becomes

$$V_0 \nabla \bar{\eta} \nabla \int^n \int h \bar{m} db = \pi_n \bar{m}.$$

For the value of a quantity at the origin, or more generally at the point belonging to the operator *outside the integral*, we shall use the suffix O whenever for clearness it seems desirable. The other end of the distance defining h is the point belonging to the element db *inside the integral*. Such a point we will denote when it seems desirable by I, and a function at I by the suffix I. Thus in the present

case, putting in O and I everywhere,

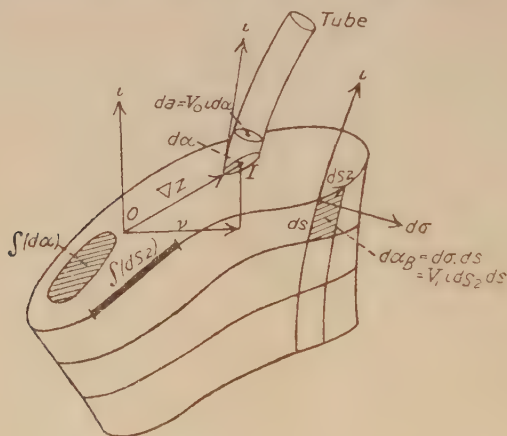
$$\begin{aligned} & V_O \nabla_O \bar{\eta}_O \nabla_O \int^n \int h_{OI} \bar{m}_I db_I = \pi_n \bar{m}_O, \\ \text{or} \quad & V_O (\nabla \bar{\eta} \nabla)_O \int^n \int h \bar{m}_I db_I = \pi_n \bar{m}_O. \end{aligned}$$

Even when we pass one or both of the symbols ∇ , or the symbol η from outside to inside, we can retain the suffix O to remind us that the symbol still *belongs* to the point O , though it has temporarily migrated to a neighbour's location.

These fairly familiar matters have been detailed by way of introduction. Let me now try to indicate what I finally succeeded in satisfying myself of, in connexion with a time-space-like manifold. I started with the full expectation that, whether I could find it or not, there was some theorem very analogous to (1), and like (1) referring to an n -dimensional region, and treating all geodesics impartially. *There is no such theorem.* Poisson's original theorem is interpretable in connexion with sources whose discharge takes place uniformly in all directions along geodesics. Now the solid angle (infinite) of all the three compartments (the inside of two semi-cones, past and future, and the spatial outside) into which the light-cone of a point partitions the manifold is in each case infinitely nearly concentrated on the cone. Hence, strengths of source being finite, the discharges when uniform in all directions, as prescribed, must be solely along the rays (*corresponding* to geodesics) of the cone. *But* there is nothing given by the fundamental form $V_o d\rho \eta d\rho$ which enables us to divide the rays up into equal bundles: we cannot do so even in a Galilean manifold. The very conception of such a partition of the rays requires us to assign a time axis; it cannot be effected absolutely. This is our first great leading; we must suppose a congruence of time-like curves filling the manifold to be given, in order to fix the time axis everywhere—that is, if we would talk about sources discharging uniformly in all directions along geodesics (or their limiting rays on the cone). Proceeding to the details of the discharge so postulated, we naturally discover that the underlying phenomenon is $(n-1)$ -dimensional, instead of n -dimensional. It is a phenomenon referring to an unbounded prescribed space-like surface (giving an $(n-1)$ -dimensional *space*), at every point of which there is a prescribed contravariant time-like vector which may be regarded as specifying the *time* axis at that point.

The last sentence truly describes the mathematical requirements for the application of (2); a given wholly

space-like region of $n-1$ dimensions, $\int (d\alpha)$, wherein [see diagram drawn for $n=3$] is situated the point O , to which the outside operator belongs, and a given unit ($V_0 \iota \eta \iota = 1$)



Vectors: ι , ∇z , v , $d\alpha$, $d\alpha_B$, $d\sigma$.

NOTE.— $ds\lambda_2$ is not a mere vector; it is a ${}_nV_2$ of $n-2$ dimensions. $d\alpha$ is a vector of $n-1$ dimensions. ds_2 is the common boundary of two intersecting regions $d\alpha$, $d\alpha_B$, and is expressible as a scalar multiple of $V_2 d\alpha d\alpha_B$.

contravariant vector ι at every point of the region. Nevertheless, it is better to consider the region as but one member of a family of the like, filling the manifold, and the contravariant vectors as the values, at the region selected, of the unit vectors specifying a whole congruence of curves throughout the manifold. The family and the congruence, one or both, or each partially, may be regarded as either prescribed or determinable at will according to the nature of the problem in hand. The space, as we may now call it, $\int (d\alpha)$ will sometimes below be taken to mean the light-cone, or rather any selected portion of it, but generally it is convenient to think of $\int (d\alpha)$ as a section of the manifold corresponding to a particular instant of coordinate time, however defined. We shall generally speak of it as the surface $t = \text{const.}$, but when this would be misleading we can say instead $f = \text{const.}$

In (2) $d\alpha$ is an anti-density covariant vector (meaning a covariant vector multiplied by k^{-1}) element of the space or region. The curves ι through the boundary of the element $d\alpha$ form an elementary tube of which much will be said as our argument proceeds. The cross-section of the tube is $kV_0 \iota d\alpha$. This in (2) is multiplied by $k^{-1}m$ as

well as by h . It is this element $V_0 \iota d\alpha$ of tubular cross-section, not the corresponding portion of the region, which is analogous to the element of volume in Poisson's original theorem; for $\bar{m}$ occurs outside the integral as well as inside, and h is the distance function connecting points O with points I . What still remains under the integral sign, namely $V_0 \iota d\alpha$, is analogous to what similarly remains in Poisson's case, the element of volume. This remark will be found true in physics generally. It is always the three-dimensional cross-section of a physical four-dimensional tube, whose generators are time-like curves, which is the mathematical representative of the physicist's three-dimensional volume. There seems no exception; certainly there is not in the domains of matter and electricity. [See my third R.S. paper on "Multenions and Differential Invariants." This paper was despatched many months before the meeting of the British Association last year (meaning the year 1921): I read that the tubes received some attention at that meeting.] It appears from the present paper that this dominance of the n -dimensional tubes is a mathematical consequence of the fundamental properties implied by η ; it is not, on the contrary, something in addition imposed by an anthropomorphic legislator of Nature: the legislator has merely said, Let the world be Riemannian and for the rest you may do as you please.

The diagram implies correctly that $d\alpha$ is a (scalar) component of the (vector) $d\mathbf{x}$ and that ν is a (vector) component of the (vector) ∇z , but is misleading as to the relative magnitudes of original and component of original. Thus actually

$$\begin{array}{ccc} \text{magnitude of } d\alpha & > & \text{magnitude of } d\alpha, \\ \text{,,} & \nu > & \text{,,} \quad \nabla z. \end{array}$$

The reader is warned to beware of misdirection from surprises of this kind, which, of course, are due to the imaginary dimension coming into comparison with the real dimensions.

As the diagram implies, it is often desirable to consider an n -dimensional region $\int (db)$ as bounded by two faces, that is, two members $\int (d\alpha)$ of the family, and by an edge consisting of generators $\int (\iota ds)$ of the tubes, ds being interval measured along ι . [$db, d\alpha, d\alpha_B, ds_2$ are all cases of ds_a , and are therefore all anti-density covariant multenions; db being a scalar, $d\alpha$ a vector, and ds_2 a ${}_nV_2$]. ds_2 is a common element of boundary of $\int (d\alpha)$ and $\int (d\alpha_B)$. $d\alpha_B$ may be taken as a parallelogram-like element of which $n-2$ sides belong to

$d\varsigma_2$ and the remaining one is ιds . It is convenient to introduce a vector $d\sigma$ parallel to $d\alpha_B$, but of the one dimension due to ds less than $d\alpha_B$. Thus

$$\left. \begin{aligned} d\alpha_B &= V_1 \iota d\varsigma_2 \cdot ds = d\sigma \cdot ds, \\ d\sigma &= V_1 \iota d\varsigma_2, \\ V_3 d\varsigma_2 d\sigma &= 0 = V_3 d\varsigma_2 d\alpha_B = V_3 d\varsigma_2 d\alpha. \end{aligned} \right\} \dots (7)$$

[Note that between two consecutive members $\int (d\alpha)$ the ds may or may not vary from tube to tube, ι , according to our prescription in the problem in hand.] $d\sigma$ is to be taken as the vector element of boundary of $\int (d\alpha)$ when we explicitly use $da(V_0 \iota d\alpha)$ as the element of integration of $\int (d\alpha)$. da is the element of physical volume of the $(n-1)$ -dimensional physical space $\int (d\alpha)$ and $d\sigma$ is the corresponding element of physical boundary; $d\sigma$ is *not* in the region $\int (d\alpha)$, but instead it is normal to the tube ι .

Our original integration theorem [*"Multenions and Differential Invariants,"* first paper, § 8, eq. (6); there are many misprints in the paper; db is omitted from the right of the equation here quoted] gives

$$\int^{n-1} \int \phi d\alpha = \int^n \int \phi_9 \nabla_9 db, \quad \dots (9)$$

but in our present notation $d\alpha$ refers only to the faces, and the present form of (9) will be

$$\int^{n-1} \int \phi d\alpha_B = - \int^{n-1} \int \phi d\alpha + \int^n \int \phi_9 \nabla_9 db. \quad \dots (10)$$

If on bringing the two faces close together, and prescribing that ds is common to all the elements of tubes between the faces (this prescribes the consecutive member $\int (d\alpha)$, when a particular one is given) we should have $db = V_0 \iota d\alpha \cdot ds$. If we could assume the two opposite ends of a tube to cause the sum $\phi d\alpha + \phi d\alpha'$ to be zero, we should have from (10)

$$(\text{incomplete}) \quad \int^{n-2} \int \phi d\sigma = \int^{n-1} \int \phi_9 \nabla_9 da, \quad \dots (11)$$

exactly similar to (9). Unfortunately for simplicity, this is legitimate only in very special cases.

The true form of (11) may be obtained from our general theorem of integration, by which in particular

$$\int^{n-1} \int \psi d\varsigma_2 = \int^{n-1} \int \psi_9 V_2 d\alpha \nabla_9. \quad \dots (12)$$

Now, by (7) $d\sigma = V_1 \iota d\varsigma_2$. Hence by (12)

$$\int^{n-2} \int \phi d\sigma = \int^{n-1} \int \phi_9 V_1 \iota_9 V_2 d\alpha \nabla_9. \quad \dots (13)$$

It will be seen from this, since $da = V_0 \iota d\alpha$, that

$$\int^{n-2} \phi d\sigma - \int^{n-1} \phi_9 \nabla_9 da$$

instead of being zero, as stated in (11), consists of two terms,

$$- \int^{n-1} \phi_9 d\alpha V_0 \iota \nabla_9 - \int^{n-1} \phi V_1 (V_2 d\alpha \nabla \cdot \iota),$$

the first involving the derivative $-V_0 \iota \nabla \cdot \phi$ of ϕ along the tube ι , and the second the derivatives of ι in the form $-V_1 (V_2 d\alpha \nabla \cdot \iota)$. This last has a simple geometrical interpretation. Integrating it over a finite portion of $\int (d\alpha)$, we have by (12)

$$- \int^{n-1} \phi V_1 (V_2 d\alpha \nabla \cdot \iota) = - \int^{n-2} \phi V_1 d\sigma_2 \iota = \int^{n-2} \phi d\sigma$$

by (7). This geometrical result reminds us of the important fact that $\int^{n-2} \phi d\sigma$, for a closed boundary, in general differs from zero. In a Euclidean space it would be zero.

Of special importance is the following particular case of (13) obtained by putting $\phi(\cdot) = V_0 \bar{\tau}(\cdot)$, where τ is a contravariant vector density:

$$\int^{n-2} \int V_0 \bar{\tau} d\sigma = \int^{n-1} \int V_0 d\alpha V_1 \nabla V_2 \bar{\tau} \iota. \quad (14)$$

§ 3. *Proof of the two forms of Poisson's theorem.*—Let the position vector ρ of a point be given by

$$\rho = \sum_{c=1}^n x_c \iota_c, \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\}, \quad (15)$$

where

$$1 = -\iota_1^2 = -\iota_2^2 = \dots = -\iota_{n-1}^2 = \mp \iota_n^2$$

the upper or lower sign of ι_n^2 being taken according as the manifold is space-like or time-space-like.

To prove (1) take the point O as origin and use equiradial coordinates ["equiradial" seems a suitable term for the coordinates used in "A new Identity affecting Relativity," etc., Phil. Mag.†]. An asterisk attached to an equation signifies that it is true when the coordinates are equiradial, and in particular when they are Galilean. Thus

$$\left. \begin{array}{l} h = (-\rho^2)^{\frac{2-n}{2}} / (n-2), \\ \nabla h = -(-\rho^2)^{-\frac{n}{2}} \rho, \\ \nabla^2 h = 0, \end{array} \right\} [h \text{ given by (1)}], \quad (16)^*$$

except for the point O itself. Draw a small sphere around the origin and consider its boundary as part of the integral boundary. With the origin thus excluded

$$\begin{aligned} 0 &= \nabla^2 \int^n \int h db = \int^n \int \nabla^2 h db, \\ &= \int^{n-1} \int V_0 d\alpha \nabla h. \end{aligned}$$

† [To appear shortly.]

[It is easy to see that $V_0 d\alpha \nabla h$ is the solid angle subtended by $d\alpha$ at O.] This result shows that the integral over an arbitrary boundary surrounding the origin is equal and opposite to that of the small sphere; or if both are taken in the same direction radially, they are equal. Now the value for the small sphere is π_n . The well-known argument of three dimensions holds, and we have that with the coordinates chosen

$$\nabla^2 \int^n h \bar{m} db = \pi_n \bar{m}. \quad . \quad . \quad . \quad (17)^*$$

This is the same as (1) in equiradial coordinates and the integral of (1) is an invariant. Hence (1) is true universally.

Similarly for (2). In the present case put $x_n = t$, and $\sum_{c=1}^{n-1} x_c t_c$, and remember that

$$\nabla = \sum_{c=1}^{n-1} t_c D_{x_c} - t_n D_t. \quad . \quad . \quad . \quad (18)$$

Take O for the origin and use equiradial coordinates with axes so chosen that at the origin $t = t_n$. Thus

$$\left. \begin{aligned} 2z &= \sum x^2 - t^2, \quad \nabla z = \rho, \quad -V_0 t \nabla z = t, \\ h &= (\sum x^2)^{\frac{3-n}{2}} / (n-3). \end{aligned} \right\}. \quad (19)^*$$

This is exactly the same form of h as we had in the proof of (1), except that here we have $n-1$, where before we had n . Hence

$$[h \text{ given by (2) and (5)}] \quad \nabla^2 h = 0, \quad . \quad . \quad . \quad (20)^*$$

except where h is infinite, that is, on the axis of t . [As a matter of fact, our surface $\int (d\alpha)$, being space-like, cannot bend round and intersect the axis in a point other than O, so that O is the only singular point.]

We now have

$$\nabla^2 \int^{n-1} h \bar{m} V_0 t d\alpha = 4\pi \bar{m}, \quad . \quad . \quad (21)^*$$

because, first from (20), we may neglect all parts of the integral except near the origin, and there we may put $t = t_n$. The matter has now been reduced exactly to the former case, except that $x_1, x_2, \dots, x_{n-1}$ replace the former $x_1, x_2, \dots, x_{n-1}, x_n$. Hence (2) is true.

At this stage the following remarks seem desirable:—

(1) If the family and congruence have not been prescribed the family may be taken as the planes $x_n = \text{const.}$ in any given system of coordinates; that is, we may identify our present *coordinate* t with our former *parameter* t . At the same time we may take t parallel everywhere to t_n , which simplifies the differentiations when we transform (2), say by integration by parts. Instead, we might take ηt parallel

everywhere to ι_n . This would make the tubes everywhere orthogonal to the surfaces.

(2) The converse of (1) in all particulars holds. If both family and congruence have been prescribed, we can choose coordinates (flatten the surfaces and straighten the tubes), such that the family again becomes $x_n = \text{const.}$ and ι is everywhere parallel to ι_n ; or instead, if only the family had been prescribed, we could have taken the tubes orthogonal to them, so making η parallel everywhere to ι_n .

(3) When the family and tubes are already prescribed, we can make $\iota = \iota_n$ everywhere, and one particular member of the family become $x_n = 0$. Of course, if that one member alone has been prescribed, we can now further take the whole family to be $x_n = \text{const.}$

We will conclude by proving Green's theorem. Let y and z be invariant functions of position. [When we require the original z of (2), (3), (4), we may use $Z = \frac{1}{2} (\text{distance})^2$.] In the space-like manifold integrate twice by parts, the following:

$$\int^a \int^b y V_0 \nabla \eta \nabla z db = \int^{n-1} \int y V_0 d\alpha \eta \nabla z - \int^n \int V_0 \nabla y \bar{\eta} \nabla z db \\ = \int^{n-1} \int V_0 d\alpha \eta (y \nabla z - z \nabla y) + \int^n \int z V_0 \nabla \bar{\eta} \nabla y db.$$

Here we might have said instead: express

$$\int^{n-1} \int V_0 d\alpha \eta (y \nabla z - z \nabla y)$$

as a volume integral. Each of these processes can be imitated in the time-space-like manifold. Perhaps the second is easier to follow. Using (14) express

$$\int^{n-2} \int V_0 d\sigma \eta (y \nabla z - z \nabla y)$$

as an $\int^{(n-1)} \int (da)$. Thus we obtain the two forms of Green's theorem, both forms applying to both species of n -manifold,

$$\left. \begin{aligned} \int^n \int (y V_0 \nabla \bar{\eta} \nabla z - z V_0 \nabla \bar{\eta} \nabla y) db \\ &= \int^{n-1} \int V_0 d\alpha \bar{\eta} (y \nabla z - z \nabla y), \\ \int^{n-1} \int V_0 d\alpha (-y V_1 \nabla V_2 \iota \bar{\eta} \nabla z + z V_1 \nabla V_2 \iota \bar{\eta} \nabla y \\ &\quad - \bar{\eta} V_1 \iota V_2 \nabla y \nabla z) \\ &= \int^{n-2} \int V_0 d\sigma \bar{\eta} (y \nabla z - z \nabla y). \end{aligned} \right\} \quad (22)$$

The three dimensional form of (22) used to be known as Green's theorem. Jeans ('Electricity and Magnetism,' § 181) seems to say that Green's theorem, properly speaking, is what (9) and (14) become when we put

$$\phi = V_0 () \bar{\tau},$$

where $\bar{\tau}$ is a contravariant vector density. In (22)

$$\tau = \bar{\eta} (y \nabla z - z \nabla y).$$

XV. *The Mechanical Forces indicated by Relativity in an Electromagnetic Field. Can their existence be demonstrated?* By ALEX. MCAULAY, M.A., Professor of Mathematics, University of Tasmania*.

§ 1. **EQUATIONS** in C.G.S. measure.—In the paper “Multenions and Differential Invariants,” III. (P. R. S.) the electromagnetic field was treated on the lines of relativity confined to a Riemann manifold, on a new and, I hope, simplified basis. The special conclusions therefrom are, in the present paper, brought to a much more satisfactory halting-stage than in the former paper.

First it is desirable to indicate (see synopsis below) the changes of notation necessary to pass from rational to C.G.S. units.

t must everywhere be replaced by ct where c is the velocity of light *in vacuo*. This has to be remembered in every differentiation with regard to t ; $\partial/\partial t$ must always be replaced by $c^{-1}\partial/\partial t$. There is one very important apparent exception to the statement that t is replaced by ct . Its cause may be thus stated. Merely to *preserve simplicity* in our mathematical expressions, $\int db$ and $\int Wdb$ are no longer to be regarded as *elements of action*. They shall instead be taken to mean (action $\times c$). Thus

$$\int f \cdot dx dy dz d(ct) = \text{action} \times c$$

and we derive the statement that

$$\int f \cdot dx dy dz dt = \text{action}.$$

This is the apparent exception referred to, since it may seem here to be exceptionally imposed that t remains unchanged instead of being replaced by ct .

$\mathbf{R}$ is written for the quaternion $xi + yj + zk$, or, according to our identifications (see § 3 below) for the multenion $x\iota_2\iota_3 + y\iota_3\iota_1 + z\iota_1\iota_2$; $\dot{\nu}$ as usual is used for $\iota_1\iota_2\iota_3\iota_4$. Thus

$$\rho = \iota_4(\dot{\nu}\mathbf{R} + ct)$$

in C.G.S. units, instead of $\iota_4(\dot{\nu}\mathbf{R} + t)$ in rational units.

In C.G.S. units the $\mathbf{H}$ of rational units must be replaced by $(4\pi)^{-1}\mathbf{H}$, but we do not require to introduce a single other 4π .

The reader is recommended to think mainly of the E.M. system. Thus we shall write (c) and (c^{-1}) for what in a completely comprehensive scheme would be written as

$$(c) = c\mu = c^{-1}K, \quad (c^{-1}) = (c)^{-1}.$$

* Communicated by the Author.

In rational units, of course, D_t is written in place of the $c^{-1}D_t$ of (7).

∇ is of dimensions L^{-1} and $\omega \cdot \nabla$ of dimensions H^{-1} . Hence from the synopsis eqs. the dimensional relations are seen to be

$$HB = LF = (f), \quad X = L^{-1}H, \quad P = LB, \quad . \quad . \quad (8)$$

where (f) is the dimensional symbol of f . The reader will easily verify from these statements that all our equations are correct dimensionally. *Note.*— HB is in ordinary theory energy per unit of three-dimensional volume. Now for simplicity we desire ω^* , $\omega^\times$ to be of the same dimensions as H , B respectively; and *also* we desire to retain the simple equation

$$\omega^\times = -\omega \cdot \nabla f.$$

This requires $\int dx dy dz d(ct) = \int db$ to be of dimensions $c \times$ action as we prescribed above. It will be seen that our prescriptions have left the dimensions of H or B absolutely arbitrary; with regard to H and B we have prescribed *only* that HB is force per unit area, that is energy per unit volume (area and volume in the usual three-dimensional sense).

§ 2. The stresses and forces in an electromagnetic field.—

In § 21 of "Mult. etc." III., in the equations (23a), (19a), and in the equations (24), (25), (26), when supplemented by the expressions (24a), (25a), (26a), a vector, normal to ι , occurs and is denoted by $\xi \cdot \iota = \iota \cdot \nabla f$. We shall here denote it by $\sigma^{\bullet \times}$. The vector can be completely evaluated in terms of ω^* , $\omega^\times$.

From eq. (26), when supplemented, we have

$$2_\theta \{ \nabla f \alpha^\times = \theta^{-1} \alpha^\times \cdot f - \theta^{-1} V_1 (V_3 \omega^\times \alpha^\times) \omega^\bullet - \theta^{-1} \sigma^{\bullet \times} \cdot V_0 \iota \alpha^\times.$$

$2_\theta \{ \nabla f - f \cdot \theta^{-1} = \phi$ is self-conjugate, so that $V_2 \xi \phi \xi = 0$. Using this we obtain at once

$$V_2 \iota \theta^{-1} \sigma^{\bullet \times} = V_2 (V_1 \xi \omega^\bullet) \theta^{-1} (V_1 \xi \omega^\times). \quad . \quad . \quad (1)$$

Since $\sigma^{\bullet \times}$ is normal to ι

$$\sigma^{\bullet \times} = V_1 \iota \theta V_2 \iota \theta^{-1} \sigma^{\bullet \times}$$

or

$$\sigma^{\bullet \times} = V_1 \iota V_2 (\theta V_1 \xi \omega^\bullet) (V_1 \xi \omega^\times). \quad . \quad . \quad (2)$$

This is the really useful form of $\sigma^{\bullet \times}$ since it gives in neutral form

$$(\text{neutral form}) \quad \sigma^{\bullet \times} = V_1 \iota V_2 \omega^\bullet \omega^\times, \quad . \quad . \quad . \quad (3)$$

but if we please we can rid eq. (2) of ζ , obtaining

$$\begin{aligned}\sigma^{\bullet \times} &= \theta V_1 (V_1 \omega^{\times} \iota) \omega^{\bullet} - V_1 (V_1 \omega^{\bullet} \theta \iota) \omega^{\times} \} \\ &= \theta V_1 \omega^{\times} V_3 \iota \omega^{\bullet} - V_1 \omega^{\bullet} V_3 \theta \iota \omega^{\times} \} \end{aligned} \quad (4)$$

In Galilean coordinates we have from (3)

$$\sigma^{\bullet \times} = V_1 [V_2 (\mathbf{BH} + \mathbf{DE}) - \dot{V} V_2 (\mathbf{DB} - \mathbf{EH})] \iota. \quad (5)$$

[In § 3 below is explained in detail how readily to effect the reductions of this § 2. We have thought this procedure would render the main argument clearer.] Thus putting

$$\iota = \iota_4 (\dot{V} \mathbf{V} + 1) / \sqrt{(1 + \mathbf{V}^2)}, \quad (6)$$

so that $\mathbf{V}$ is the velocity, we obtain

$$\begin{aligned}\sigma^{\bullet \times} \sqrt{(1 + \mathbf{V}^2)} &= \iota_4 \{ \dot{V} [V (\mathbf{BH} + \mathbf{DE}) \cdot \mathbf{V}] \\ &\quad + V [\mathbf{DB} - \mathbf{EH}] \cdot \mathbf{V} \} - S (\mathbf{DB} - \mathbf{EH}) \mathbf{V}. \end{aligned} \quad (7)$$

Eq. (19 a) reads

$$-l T_{(5)} \zeta = V_1 \kappa^{\bullet \times} \omega^{\times} - \nabla_9 V_0 \iota_9 \sigma^{\bullet \times} - \sigma_9^{\bullet \times} V_0 \iota_9 \nabla_9. \quad (8)$$

Our most important deduction from this happens to be easy; so before treating (8) generally we will make this deduction. Let the coordinates be chosen so that $\mathbf{V}$ is zero at the point considered and therefore ι becomes ι_4 . In the neighbourhood of the point let the medium also be at rest; in other words let the point be in a rigid body (or, at any rate, a body wholly at rest). Thus ι is the constant ι_4 and all its derivatives are zero; and $-V_0 \iota \nabla = D_t$ and $\mathbf{V}$ of (7) is zero. Thus the quaternion expression for the force per unit volume is

$$\mathbf{VKB} - \mathbf{ES} \nabla \mathbf{D} + D_t \mathbf{V} (\mathbf{DB} - \mathbf{EH}). \quad (9)$$

The third term expresses a hitherto unsuspected (I believe) mechanical force exerted by the field. Can any physicist verify its existence? In C.G.S. measure the term is

$$D_t \mathbf{V} (\mathbf{DB} - \mathbf{EH} / 4\pi c^2).$$

[This is a case of the remark that to pass to C.G.S. units D_t must be replaced by $c^{-1} D_t$.]

Returning to (8), since $V_0 \iota \sigma^{\bullet \times} = 0$,

$$-\nabla_9 V_0 \iota_9 \sigma^{\bullet \times} = \nabla_9 V_0 \iota \sigma_9^{\bullet \times}.$$

Hence putting $\sigma^{\bullet \times} = l \sigma^{\times}$ and $l \iota = \iota$ (8) may be easily written in the invariantive form

$$-l T_{(5)} \zeta = V_1 \kappa^{\bullet \times} \omega^{\times} + V_1 (V_2 \nabla \sigma^{\times}) \iota^{\bullet} - \sigma^{\times} V_0 \nabla \iota^{\bullet}. \quad (10)$$

Each of the two terms in $\sigma^\times$ is separately normal to ι . That this portion of the force is normal to ι means that the corresponding portion of $F_4 (= -SEK)$ is zero when the medium is taken to be at rest; that is, there is no unsuspected energy transformation of an electrical or heat generating nature, due to the new term in the mechanical force.

In interpreting (10) assume, at the point considered, $\mathbf{V}$ to be zero, but neither $D_t\mathbf{V}$ nor $(\nabla_9, \mathbf{V}_9)$ to be zero. The middle term of (10) thus contributes the third term of (9) together with

$$\text{where } \left. \begin{aligned} & \mathbf{V}[\mathbf{V}(\mathbf{B}\mathbf{H} + \mathbf{D}\mathbf{E})] D_t\mathbf{V}' - \nabla_9 S(\mathbf{D}\mathbf{B} - \mathbf{E}\mathbf{H}) \mathbf{V}_9', \\ & \mathbf{V}' = \mathbf{V}/\sqrt{(1 + \mathbf{V}^2)} \end{aligned} \right\} \quad (9a)$$

The third term of (10) contributes

$$\mathbf{V}(\mathbf{D}\mathbf{B} - \mathbf{E}\mathbf{H}) \cdot (-S\nabla\mathbf{V}' + D_t[1/\sqrt{(1 + \mathbf{V}^2)}]). \quad (9b)$$

In changing to C.G.S. units $\mathbf{V}$ must, of course, be replaced by $c^{-1}\mathbf{V}$.

I imagine these terms depending on acceleration and time-rate of strain are much too small to be determined experimentally. The physicist who understands the meaning of three-dimensional vectors has been placed by the expressions above in a position to judge for himself.

In addition to the three terms (9), (9a), (9b), expressing the mechanical force when the medium is at rest there will, of course, be terms due to the coordinates of strain entering the expression for action. What has been established is that a *perfectly arbitrary* invariant function $f(\omega^*, \varphi^*, \theta)$ occurring as the field terms in W (action per unit of four-dimensional volume), a *quite definite* force per unit volume (from which the arbitrary function has completely vanished) must exist. The function, of course, has its effects, but these are merely to fix the details of interrelation of $\mathbf{B}, \mathbf{H}, \mathbf{D}, \mathbf{E}$. This result seems very surprising. Perhaps relativity will give similar guidance when elasticity etc. has been brought under its treatment.

§ 3. *Transformation from multenion to quaternion forms.*—With our identifications, a real quaternion Q is taken to mean

$$\begin{aligned} Q &= a\iota_2\iota_3 + b\iota_3\iota_1 + c\iota_1\iota_2 + g \\ &= ai + bj + ck + g = \mathbf{G} + g, \end{aligned}$$

and an imaginary quaternion $\dot{\nu}Q'$ to mean ($\dot{\nu}=\iota_1\iota_2\iota_3\iota_4$)

$$\begin{aligned}\dot{\nu}Q' &= a'\iota_4\iota_1 + b'\iota_4\iota_2 + c'\iota_4\iota_3 + g'\iota_1\iota_2\iota_3\iota_4 \\ &= \dot{\nu}(a'\iota_2\iota_3 + b'\iota_3\iota_1 + c'\iota_1\iota_2 + g') = \dot{\nu}(\mathbf{G}' + g'),\end{aligned}$$

$Q + \dot{\nu}Q'$ is a bi-quaternion (in Hamilton's sense). The general multenion q is expressed by means of two bi-quaternions $Q + \dot{\nu}Q'$, $P + \dot{\nu}P'$ and ι_4 thus

$$q = (Q + \dot{\nu}Q') + \iota_4(P + \dot{\nu}P').$$

Let a be an arbitrary real scalar and $\mathbf{A}, \mathbf{B}$ arbitrary real vectors (of the quaternion system). Then we have the following scheme :—

Forms of $V_c q$; $c=0, 1, 2, 3, 4$.

$$\begin{array}{lll}V_0 q = a & V_1 q = \iota_4(\dot{\nu}\mathbf{A} + a) & V_2 q = \mathbf{A} + \dot{\nu}\mathbf{B} \\ V_4 q = \dot{\nu}a & V_3 q = \iota_4(\mathbf{A} + \dot{\nu}a) & \end{array}$$

It is easy to pick out the five parts of q given by the last equation. Thus

$$\begin{aligned}V_0 q &= S(Q), & V_1 q &= \iota_4(\dot{\nu}VP' + SP), & V_2 q &= V(Q + \dot{\nu}Q'), \\ V_4 q &= \dot{\nu}SQ', & V_3 q &= \iota_4(VP + \dot{\nu}SP').\end{aligned}$$

The one biquaternion furnishes the even parts and the other (the ι_4 biquaternion) the odd parts, thus

$$\begin{aligned}Q + \dot{\nu}Q' &= (V_0 + V_2 + V_4)q, \\ \iota_4(P + \dot{\nu}P') &= (V_1 + V_3)q.\end{aligned}$$

The imaginary $\dot{\nu}$ and also ι_4 are both commutative with the quaternions, but they are anti-commutative with one another, or

$$\iota_4\dot{\nu} = -\dot{\nu}\iota_4.$$

As a general rule collect all the ι_i 's at the left of a complicated quadri-quaternion expression, attending to change of sign at each passage of ι_i past $\dot{\nu}$. Then use the scheme for further treatment.

These remarks receive illustration in the transformations of § 2.

$$V_1 \kappa \cdot \omega^\times = V_1 \iota_4(\dot{\nu}\mathbf{K} + K_4)(\mathbf{B} + \dot{\nu}\mathbf{E}).$$

Consulting the scheme under V_1 we have from the bi-quaternion $(\dot{\nu}\mathbf{K} + K_4)(\mathbf{B} + \dot{\nu}\mathbf{E})$ to pick out the part of the

form $\dot{\mathbf{v}}\mathbf{A} + a$, that is,

$$\dot{\mathbf{v}}(\mathbf{VKB} + \mathbf{K}_4\mathbf{E}) - \mathbf{SKE}.$$

The rest of the bi-quaternion merely gives $\mathbf{V}_3\kappa^\bullet\omega^\times$ with which we have no concern. Thus

$$\mathbf{V}_1\kappa^\bullet\omega^\times = \iota_4(\dot{\mathbf{v}}[\mathbf{VKB} + \mathbf{K}_4\mathbf{E}] - \mathbf{SKE}).$$

Hence we have the first two terms $\mathbf{VKB} + \mathbf{K}_4\mathbf{E}$ of (9).

Next deduce (5) from (3)

$$\begin{aligned}\sigma^{\bullet\times} &= \mathbf{V}_1(\mathbf{V}_2\omega^\times\omega^\bullet)\iota_4, \\ &= \mathbf{V}_1[\mathbf{V}_2(\mathbf{B} + \dot{\mathbf{v}}\mathbf{E})(\mathbf{H} + \dot{\mathbf{v}}\mathbf{D})]\iota_4, \\ &= \mathbf{V}_1[\mathbf{V}_2(\mathbf{BH} + \mathbf{DE}) - \dot{\mathbf{v}}\mathbf{V}_2(\mathbf{DB} - \mathbf{EH})]\iota_4.\end{aligned}$$

Next deduce (7) using the abbreviations

$$\begin{aligned}\mathbf{V}' &= \mathbf{V}/\sqrt{(1 + \mathbf{V}^2)}, \quad \mathbf{X} = \mathbf{V}(\mathbf{BH} + \mathbf{DE}), \quad \mathbf{Y} = \mathbf{V}(\mathbf{DB} - \mathbf{EH}), \\ 1/\sqrt{(1 + \mathbf{V}^2)} &= b,\end{aligned}$$

so that b is what is usually denoted by β . We have

$$\begin{aligned}\iota &= \iota_4(\dot{\mathbf{v}}\mathbf{V}' + b), \\ \sigma^{\bullet\times} &= \mathbf{V}_1(\mathbf{X} - \dot{\mathbf{v}}\mathbf{Y})\iota_4(\dot{\mathbf{v}}\mathbf{V}' + b), \\ &= \mathbf{V}_1\iota_4(\mathbf{X} + \dot{\mathbf{v}}\mathbf{Y})(\dot{\mathbf{v}}\mathbf{V}' + b).\end{aligned}$$

From the biquaternion () () following ι_1 we have to pick out the part of form $\dot{\mathbf{v}}\mathbf{A} + a$. Thus

$$\sigma^{\bullet\times} = \iota_1[\dot{\mathbf{v}}(\mathbf{VXV}' + b\mathbf{Y}) - \mathbf{SYV}']$$

which is eq. (7).

Similarly (11) and (12) can be deduced from (10).

In conclusion it may be remarked that

$$a\iota_1 + b\dot{\mathbf{v}} + c\iota_1\dot{\mathbf{v}} + g$$

is itself a quaternion such that we may put

$$\iota_4 = i\sqrt{(-1)}, \quad \dot{\mathbf{v}} = j, \quad \iota_4\dot{\mathbf{v}} = k\sqrt{(-1)}.$$

University of Tasmania,
28 November, 1922.